Model type
antiGen
antiGen Finetunes
Deep Mutational Scans
Pandemic Data in Training Window
Pre−Pandemic Data Only
Ensembles
Val−selected, averaged over 10 holdouts

A
Recall, all list cells, cell−weighted [20 AA, revised corpus]
P(listed mutation in top q)
Quantile (q)
Model
antiGen
antiGen as ESM 2 (35M) LM head FT
BA.1 DMS (expression)
Where × which
EVEscape
Mean of z−scores
antiGen, 10x val−selected epoch (mean of log−probs)

B
Recall, de novo, cell−weighted [20 AA, revised corpus]
Quantile (q)
Model
antiGen
antiGen as ESM 2 (35M) LM head FT
BA.2 DMS (expression)
Where × which
EVEscape
Mean of log−probs
antiGen, 10x val−selected epoch (mean of log−probs)

C
Top decile, all list cells, cell−weighted [20 AA, revised corpus]
antiGen
antiGen as EVE full FT
antiGen as Tranception full FT
antiGen as ESM 2 (35M) partial FT
antiGen as ESM 2 (35M) LM head FT
antiGen as ESM 2 (35M) full FT
antiGen as ESM 2 (650M) partial FT
antiGen as ESM 2 (650M) LM head FT
antiGen as ESM 2 (650M) full FT
PSSM
PSSM Conservation
MSA Transformer
MSA Trans. (BA.2 eval)
MSA Trans. (XFG eval)
EVE
Tranception
Substitution matrix
Where × which
BA.1 DMS (bind)
BA.1 DMS (expression)
BA.2 DMS (bind)
BA.2 DMS (expression)
EVEscape
ESM 2 (35M), untrained
ESM 2 (650M), untrained
BLOSUM62
Codon table
Mean of ranks
Mean of log−probs
Mean of z−scores
antiGen, 10x val−selected epoch (mean of log−probs)
Where x which + PSSM hybrid, 10x val−selected blend
2nd
3rd
1st
Recall at q = 0.1

D
Top decile, de novo, cell−weighted [20 AA, revised corpus
2nd
3rd
1st
Recall at q = 0.1

E
AUC, all list cells, cell−weighted [20 AA, revised corpus]
antiGen
antiGen as EVE full FT
antiGen as Tranception full FT
antiGen as ESM 2 (35M) partial FT
antiGen as ESM 2 (35M) LM head FT
antiGen as ESM 2 (35M) full FT
antiGen as ESM 2 (650M) partial FT
antiGen as ESM 2 (650M) LM head FT
antiGen as ESM 2 (650M) full FT
PSSM
PSSM Conservation
MSA Transformer
MSA Trans. (BA.2 eval)
MSA Trans. (XFG eval)
EVE
Tranception
Substitution matrix
Where × which
BA.1 DMS (bind)
BA.1 DMS (expression)
BA.2 DMS (bind)
BA.2 DMS (expression)
EVEscape
ESM 2 (35M), untrained
ESM 2 (650M), untrained
BLOSUM62
Codon table
Mean of ranks
Mean of log−probs
Mean of z−scores
antiGen, 10x val−selected epoch (mean of log−probs)
Where x which + PSSM hybrid, 10x val−selected blend
3rd
2nd
1st
Recall AUC

F
AUC, de novo, cell−weighted [20 AA, revised corpus]
3rd
1st
2nd
Recall AUC